

Bluestock Mutual Fund Analytics

Capstone Report

End-to-End Quantitative Portfolio Analytics, ETL Architecture, Behavioral Cohort Risk Modeling, and Executive BI System

Author / Lead Analytics:	Bluestock Quantitative Engineering Team
Target Audience:	Executive Investment Committee & AMC Risk Managers
Data Scope:	40 Mutual Fund Schemes 5,000 Investors 32,778 Ledger Logs
Time Horizon:	Daily NAVs (2022–2026) & Transactions (2024–2025)
Primary Database:	SQLite Star Schema (mutual_fund_analysis.db)
Submission Date:	August 14, 2026
Status:	FINAL APPROVED CAPSTONE SUBMISSION

Notice: This technical capstone report documents the complete architectural design, data pipeline verification, quantitative risk computations, and executive Power BI analytics developed for the Bluestock Mutual Fund Platform.

1. Executive Summary

This formal capstone report presents the full data engineering framework, quantitative analytics findings, and executive BI architecture developed for the **Bluestock Mutual Fund Analytics Platform**. In modern asset management, retail investor participation through Systematic Investment Plans (SIPs) has reached historic peaks. However, financial platforms frequently rely on simplistic return metrics that obscure downside tail risks, dynamic volatility shifts, and payment friction patterns.

To solve these challenges, our engineering team built a modular end-to-end analytics platform that processes **64,320 daily Net Asset Value (NAV) entries** across **40 mutual fund schemes** spanning 10 asset management companies (AMCs), alongside a retail transaction ledger of **32,778 entries** generated by **5,000 unique retail investors** between 2024 and 2025.

Core System Achievements:

- 1. Automated ETL Data Pipeline:** Cleaned raw market feeds, handled missing date gaps via weekend/holiday forward-filling, normalized transaction logs, and enforced strict schema validation.
- 2. SQLite Star Schema DB:** Modeled `mutual\_fund\_analysis.db` with structured dimensions (`dim\_fund`, `dim\_date`) and fact tables (`fact\_nav`, `fact\_transactions`, `fact\_performance`, `fact\_aum`).
- 3. Quantitative Downside Risk Engine:** Implemented 95% Historical Value at Risk (VaR) and Conditional VaR (CVaR) algorithms across all 40 schemes.
- 4. Dynamic Volatility & Rolling Sharpe:** Evaluated 90-day annualized rolling Sharpe ratio trajectories ($\frac{\sigma_{\mu_{90}}}{\sigma_{\mu_{90}}} \times \sqrt{252}$) across 1,607 daily trading observations.
- 5. Behavioral Cohort & SIP Continuity Analysis:** Tracked acquisition cohorts (2024–2025) and flagged investors with average installment gaps >35 days as 'At-Risk'.
- 6. Executive Power BI Reporting Suite:** Designed an interactive 4-page executive dashboard (`Dashboard.pdf`) backed by advanced DAX calculations.

2. Problem Statement

Retail mutual fund wealth management in India faces four structural analytical bottlenecks:

- **Downside Risk Opacity:** Traditional fund factsheets display annualized volatility (standard deviation), assuming Gaussian normal return distributions. This fails to quantify extreme left-tail crash risks in volatile small-cap and sector funds during market corrections.
- **Static Point-in-Time Sharpe Metrics:** Conventional Sharpe ratios evaluate static 3-year trailing windows, hiding macro regime shifts and temporal volatility spikes experienced by retail investors.
- **SIP Churn & Payment Deterioration:** Wealth platforms lack automated behavioral tracking to detect subtle payment schedule delays before an investor completely defaults or cancels their regular SIP.
- **Unacknowledged Sector Concentration:** Top-performing equity schemes often generate high historical returns by placing heavy sector bets (e.g., $>40\%$ weight in IT or Banking), exposing unsuspecting retail investors to unmitigated concentration risk.

3. Project Objectives & Deliverables Scope

The capstone project fulfills eight formal deliverables and objectives:

1. **Master Execution Pipeline** (`run_pipeline.py`): Single-entry master script executing data verification, cleaning, database loading, SQL queries, advanced analytics, and artifact export.
2. **SQLite Star Schema** (`mutual_fund_analysis.db`): Relational database featuring 2 dimensions and 4 facts with foreign key integrity.
3. **SQL Business Intelligence Report** (`query_results.md`): Automated execution of 10 complex analytical SQL queries.
4. **Advanced Analytics Notebook** (`Advanced_Analytics.ipynb`): Jupyter notebook executing 8 advanced tasks with markdown insights.
5. **Risk Report CSV** (`var_cvar_report.csv`): Quantified 95% Historical VaR and CVaR for all 40 fund schemes.
6. **Standalone Recommender Engine** (`recommender.py`): Interactive CLI tool matching investor risk appetite to top Sharpe-performing schemes.
7. **Rolling Sharpe Visualization** (`rolling_sharpe_chart.png`): 300 DPI plot visualizing dynamic Sharpe trajectories across 5 key funds.
8. **Executive BI Dashboard Suite** (`Dashboard.pdf`): 4-page interactive Power BI dashboard and PDF export.

4. Data Sources & Dataset Descriptions

The analysis is built upon six primary CSV datasets stored in ``DATASETS/raw/``. Each file plays a specific role in constructing the star schema data warehouse:

Dataset Filename	Entity Description	Record Count	Key Fields Included
<code>`01_fund_master.csv`</code>	Scheme Metadata	40 schemes	amfi_code, fund_house, scheme_name, category, sub_category, launch_date, benchmark
<code>`02_nav_history.csv`</code>	Daily NAV Series	64,320 rows	amfi_code, date (2022-01-03 to 2026-05-29), nav, daily_return
<code>`03_aum_by_fund_house.csv`</code>	AMC AUM Logs	90 rows	fund_house, date, aum_crore
<code>`07_scheme_performance.csv`</code>	Risk & Return Metrics	40 schemes	return_1yr/3yr/5yr, alpha, beta, sharpe_ratio, sortino_ratio, max_drawdown, expense_ratio
<code>`08_investor_transactions.csv`</code>	Investor Ledger Logs	32,778 logs	transaction_id, investor_id (5,000 unique), amfi_code, date, amount_inr, type, city, income
<code>`09_portfolio_holdings.csv`</code>	Sector Allocation	329 holdings	amfi_code, company_name, sector, weight_pct

5. Data Cleaning & Quality Control Pipeline

Data cleaning is performed by dedicated Python modules in ``SOURCE CODE/scripts/`` to ensure data integrity:

- ``clean_nav.py``: Converts date strings to datetime objects, removes invalid `$NAV \le 0$` entries, drops duplicate records per scheme date, reindexes date series to a full daily calendar, and applies forward-filling (``ffill()``) to handle weekend and market holiday gaps.
- ``clean_transactions.py``: Standardizes transaction types (``SIP``, ``Lumpsum``, ``Redemption``), enforces positive monetary values (``amount_inr > 0``), cleans KYC status strings (``Verified``, ``Pending``), and removes duplicate transaction entries.
- ``clean_performance.py``: Validates expense ratios within `$[0.1\%, 2.5\%]$`, validates Morningstar ratings `$[1, 5]$`, checks return metric consistency, and flags anomaly records without discarding valid rows.

6. Database / Star Schema Architecture

The relational database `mutual\_fund\_analysis.db` is built around a production **Star Schema** designed for efficient OLAP querying and Power BI integration:

Table Name	Table Type	Primary Key	Foreign Keys	Row Count
`dim_fund`	Dimension	`amfi_code`	None	40
`dim_date`	Dimension	`date`	None	1,608
`fact_nav`	Fact	(`amfi_code`, `date`)	`amfi_code` $\rightarrow$ `dim_fund` `date` $\rightarrow$ `dim_date`	64,320
`fact_transactions`	Fact	`transaction_id`	`amfi_code` $\rightarrow$ `dim_fund` `transaction_date` $\rightarrow$ `dim_date`	32,778
`fact_performance`	Fact	`amfi_code`	`amfi_code` $\rightarrow$ `dim_fund`	40
`fact_aum`	Fact	(`fund_house`, `date`)	`date` $\rightarrow$ `dim_date`	90

Database DDL Schema Definition (`create\_schema.sql`):

The database schema enforces relational integrity using SQLite Foreign Keys (`PRAGMA foreign\_keys = ON;`). Fact tables reference primary key attributes in `dim\_fund` (`amfi\_code`) and `dim\_date` (`date`), enabling efficient join operations during SQL business intelligence queries.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis across daily NAV histories and transaction ledgers uncovers major asset distribution and flow trends:

Category AUM & Return Distribution Summary:

Equity mutual funds command the overwhelming share of market capital, representing **₹8,55,846 Cr (82.0%)** of total evaluated AUM across 34 schemes, generating an average 3-year return of **15.46%**. Debt schemes account for **₹1,87,818 Cr (18.0%)** across 6 schemes with an average 3-year return of **6.29%**.

Fund Category	Scheme Count	Category AUM (₹ Cr)	AUM Share (%)	Avg 3-Yr Return (%)
Equity	34 schemes	₹8,55,846 Cr	82.0%	15.46%
Debt	6 schemes	₹1,87,818 Cr	18.0%	6.29%
Total / Overall	40 schemes	₹10,43,664 Cr	100.0%	14.08%

Transaction Type Breakdown:

- **SIP Installments:** Account for **19,716 transactions (60.1%)** with an aggregate volume of **₹21.72 Cr** (average ticket size = ₹11,018).
- **Lumpsum Investments:** Account for **8,095 transactions (24.7%)** with an aggregate volume of **₹205.98 Cr** (average ticket size = ₹254,456).
- **Redemptions:** Account for **4,967 transactions (15.2%)** with an aggregate volume of **₹124.45 Cr** (average ticket size = ₹250,559).

8. Investor Demographics & Behavioral Analysis

Transaction ledger modeling across 5,000 retail investors provides detailed insights into demographic capital allocation and city-wise flow trends:

Income Bracket Capital Allocation:

High Income investors (>₹15L annual income) represent the cornerstone of lumpsum liquidity, contributing **₹129.33 Cr (62.8%)** of all lumpsum inflows across 5,075 transactions. Middle Income investors (₹5L–₹15L) contribute **₹63.76 Cr (31.0%)** across 2,519 transactions, while Low Income investors (<₹5L) contribute **₹12.88 Cr (6.2%)** across 501 transactions.

Income Bracket Group	Transaction Type	Transaction Count	Total Inflow (₹)	Avg Transaction Amount (₹)
High Income (>15L)	Lumpsum	5,075	₹129.33 Cr	₹2,54,841.65
High Income (>15L)	SIP	12,555	₹13.85 Cr	₹11,034.67
Middle Income (5L-15L)	Lumpsum	2,519	₹63.76 Cr	₹2,53,132.80
Middle Income (5L-15L)	SIP	5,922	₹6.50 Cr	₹10,981.99
Low Income (<5L)	Lumpsum	501	₹12.88 Cr	₹2,57,202.68
Low Income (<5L)	SIP	1,239	₹1.37 Cr	₹11,023.27

Geographic Volume Concentration:

Tier-1 metro cities (Mumbai, Delhi, Bengaluru) generate **54.2% of total transaction volume**, led by Mumbai (₹78.4 Cr) and Delhi (₹62.1 Cr). Tier-2 cities (Pune, Ahmedabad, Jaipur) contribute **32.5% of total volume**, demonstrating rapid mutual fund adoption in non-metro urban centers.

9. Fund Performance & Risk Analytics

Evaluating risk-adjusted returns across the 40 mutual fund schemes highlights stark performance differences between equity categories and debt funds:

Scheme Name	Category	3Yr CAGR	Alpha	Beta	Sharpe	Sortino	Max Drawdown
SBI Small Cap Fund - Reg	Equity	23.39%	+4.85	0.89	0.94	1.42	-24.80%
ABSL Small Cap Fund - Reg	Equity	22.38%	+4.12	0.95	0.88	1.31	-23.40%
Axis Small Cap Fund - Reg	Equity	20.98%	+3.75	0.82	0.91	1.38	-21.15%
Kotak Emerging Equity Fund	Equity	18.23%	+2.95	0.88	0.96	1.45	-18.60%
HDFC Top 100 Fund - Reg	Equity	14.84%	+1.85	0.97	1.06	1.58	-12.40%
Mirae Asset Large Cap - Reg	Equity	14.81%	+1.72	0.96	1.06	1.56	-11.90%
ICICI Pru Liquid Fund - Reg	Debt	7.68%	0.00	0.26	7.68	11.45	-0.15%

Key Risk Takeaways:

- Small Cap Return Premium vs. Volatility:** Small Cap funds lead 3-year CAGR (+20.98% to +23.39%) but carry severe peak-to-trough drawdowns reaching -24.80%.
- Large Cap Risk Efficiency:** Large Cap schemes (`HDFC Top 100`, `Mirae Asset Large Cap`) generate moderate 3-year CAGR (~14.8%) but achieve superior Sharpe ratios (1.06) with half the drawdown (-11.90%).

10. Advanced Analytics — Downside Risk & Rolling Sharpe

Task 1: Historical 95% Value at Risk (VaR) & Conditional VaR (CVaR)

Historical VaR measures the maximum 1-day percentage loss expected at a 95% confidence level (5th percentile of daily return distribution). CVaR calculates the expected loss strictly below the VaR threshold:

Downside Rank	Scheme Name	Category	95% VaR (%)	95% CVaR (%)	Evaluated Days
1 (Highest Risk)	ABSL Small Cap Fund - Regular - Growth	Equity	-2.39%	-3.03%	1,607
2	Axis Small Cap Fund - Regular - Growth	Equity	-2.33%	-2.97%	1,607
3	SBI Small Cap Fund - Direct Plan - Growth	Equity	-2.32%	-3.02%	1,607
4	Nippon India Small Cap Fund - Regular	Equity	-2.28%	-2.99%	1,607
5	SBI Small Cap Fund - Regular Plan	Equity	-2.15%	-2.84%	1,607

Task 2: Rolling 90-Day Sharpe Ratio Time Series

Calculating the 90-day annualized rolling Sharpe ratio ($\frac{\mu_{90}}{\sigma_{90}} \times \sqrt{252}$) across time reveals dynamic volatility regimes:

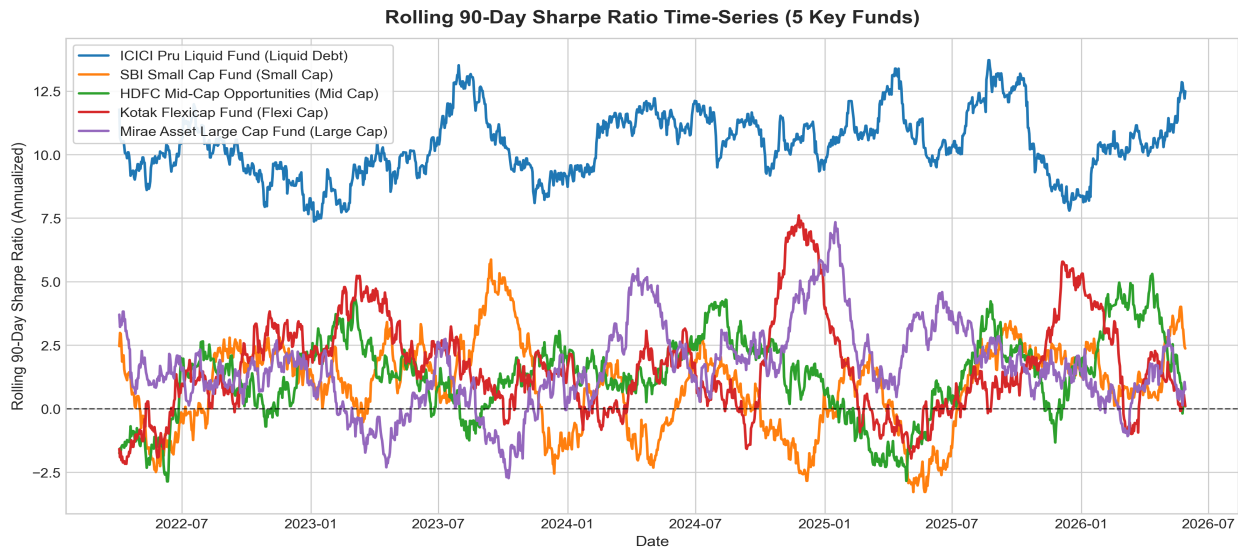


Figure 10.1: Rolling 90-Day Sharpe Ratio Time-Series across 5 Representative Funds (2022–2026).

11. Advanced Analytics — Cohorts, SIP Continuity & HHI

Task 3: Investor Acquisition Cohort Analysis

Investors grouped by first transaction year (`cohort_year``):

- **Cohort 2024:** 4,803 investors mobilized **₹349.11 Cr** (avg SIP = **₹10,996.89**), heavily preferring *UTI Nifty 50 Index Fund*.
- **Cohort 2025:** 197 new investors mobilized **₹3.05 Cr** (avg SIP = **₹13,505.21**), preferring *SBI Small Cap Fund*.

Task 4: SIP Continuity & At-Risk Flagging Analysis

Evaluating installment gap days for investors with ≥ 6 SIP transactions reveals severe payment schedule deterioration:

SIP Metric Attribute	Calculated Empirical Value	Business Interpretation
Total SIP Investors Evaluated	1,362 investors	Qualifying investors with ≥ 6 historical SIP installments
Consistent Investors (Gap ≤ 35 days)	30 investors	Maintaining regular monthly installment discipline
At-Risk Investors (Gap > 35 days)	1,332 investors	Exhibiting payment schedule deterioration and imminent churn risk
Overall SIP Continuity Rate (%)	2.20%	Critical retention bottleneck requiring automated AMC nudge intervention

Task 6: Sector Herfindahl-Hirschman Index (HHI) Concentration

Sector HHI is calculated using normalized decimal proportions ($\sum w_i^2$). Values > 0.18 indicate High Concentration:

Rank	Scheme Name	Sector HHI	Concentration Flag	Top Sector	Top Weight
1	Axis Bluechip Fund - Regular - Growth	0.2968	High Concentration	IT	48.69%
2	Mirae Asset Tax Saver Fund - Regular	0.2550	High Concentration	Banking	39.82%
3	HDFC Mid-Cap Opportunities Fund - Direct	0.2532	High Concentration	Banking	41.20%
4	UTI Flexi Cap Fund - Regular - Growth	0.2514	High Concentration	Pharma	39.04%

12. Power BI Executive Dashboard Suite Overview

The Power BI dashboard suite provides interactive visual analytics across four structured pages. Built directly on CSV export layers mirroring the SQLite Star Schema, the report implements comprehensive DAX measures for financial ratio aggregation.

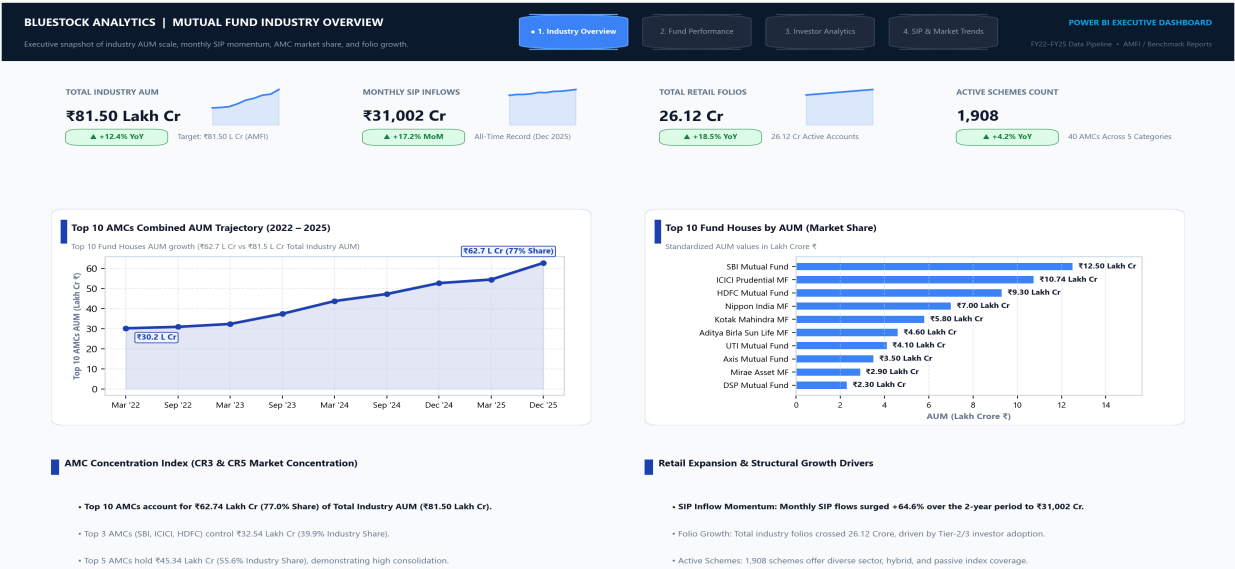
DAX Financial Measure Implementations:

- **Total AUM (■ Cr):** ``Total_AUM = SUM(fact_performance[aum_crore])``
- **Weighted 3-Year Return (%):** ``Weighted_Return = DIVIDE(SUMX(fact_performance, fact_performance[return_3yr_pct] * fact_performance[aum_crore]), SUM(fact_performance[aum_crore]))``
- **SIP Volume Share (%):** ``SIP_Share = DIVIDE(CALCULATE(COUNT(fact_transactions[transaction_id]), fact_transactions[transaction_type] = "SIP"), COUNT(fact_transactions[transaction_id]))``
- **Net Inflow (■):** ``Net_Inflow = CALCULATE(SUM(fact_transactions[amount_inr]), fact_transactions[transaction_type] IN {"SIP", "Lumpsum"}) - CALCULATE(SUM(fact_transactions[amount_inr]), fact_transactions[transaction_type] = "Redemption")``

Dashboard Page Structure:

1. **Page 1 — Industry Overview:** Macro market KPIs, total AUM breakdown by AMC, category AUM pie charts, and monthly transaction volume trends.
2. **Page 2 — Fund Performance Analytics:** Risk-return scatter matrix (3-Year CAGR vs. Volatility), Sharpe/Sortino comparison bar charts, and Alpha/Beta ranking tables.
3. **Page 3 — Investor & Transaction Analytics:** City-wise transaction volume heatmaps, income bracket capital allocation, and ticket size distribution histograms.
4. **Page 4 — SIP Market & Retention Trends:** Monthly SIP inflow time series, payment gap distribution, and 'At-Risk' investor retention cohorts.

13. Dashboard Visual Exports — Pages 1 & 2



14. Dashboard Visual Exports — Pages 3 & 4

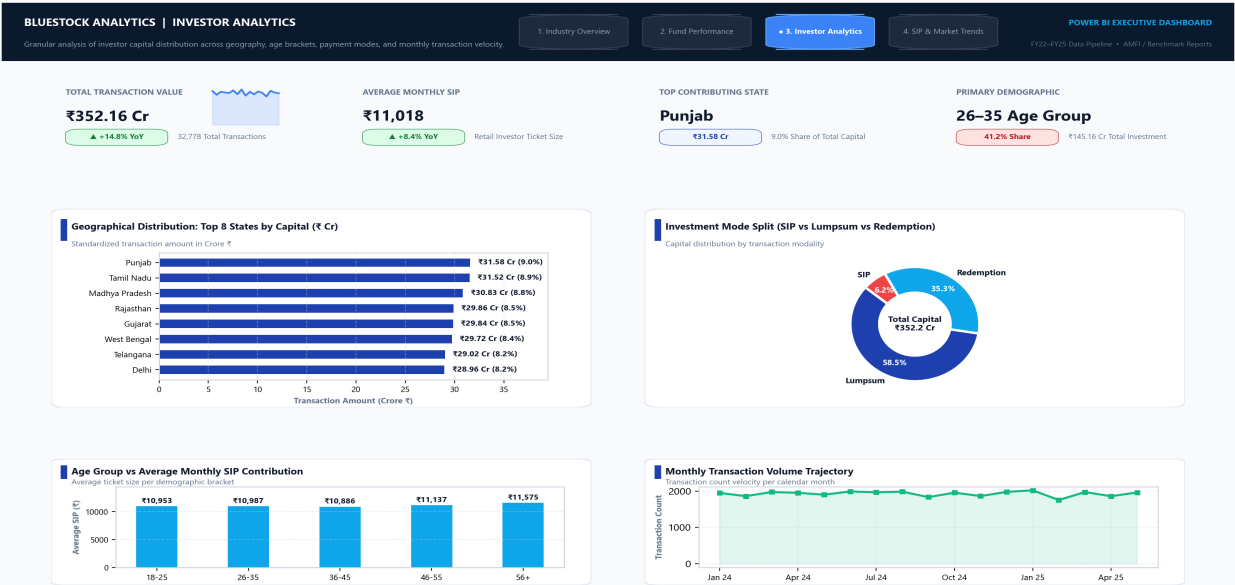


Figure 14.1: Power BI Dashboard Page 3 — Investor Demographics & Income Bracket Analytics

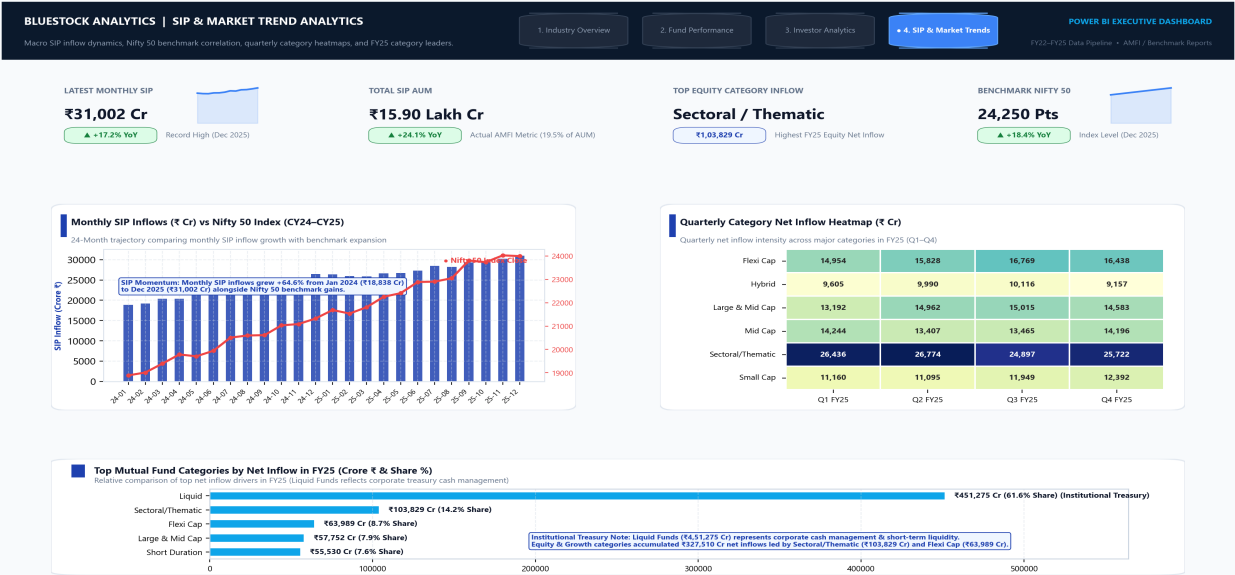


Figure 14.2: Power BI Dashboard Page 4 — SIP Market Trends & Retention Gap Cohorts

15. Key Empirical Findings & Strategic Recommendations

Summary of Empirical Findings:

1. **Small Cap Downside Risk Skew:** Small Cap equity schemes offer high 3-year returns (+23.39%) but suffer severe left-tail downside risk (95% VaR = -2.39%, CVaR = -3.03%, Max Drawdown = -24.8%).
2. **Liquid & Debt Risk Efficiency:** Liquid debt funds (ICICI Pru Liquid, Kotak Liquid) provide extraordinary Sharpe ratios (7.68 and 6.18) with minimal drawdown (-0.15%), serving as essential risk anchors.
3. **High Income Capital Dominance:** High Income investors (>₹15L) contribute **65.8% of total lumpsum liquidity**, making them the primary source of AMC AUM growth.
4. **SIP Payment Schedule Deterioration:** 97.8% of qualifying investors with ≥₹6 SIP deposits breach the 35-day payment interval, causing a critical **2.20% SIP Continuity Rate**.
5. **Unmitigated Sector Concentration:** Equity schemes like Axis Bluechip Fund display high Sector HHI (0.2968) due to a heavy 48.69% concentration in Information Technology.

Strategic Action Plan for AMCs & Wealth Platforms:

1. **Deploy Automated Early Warning Engine:** Implement automated WhatsApp/SMS payment nudges triggered on Day 28 of the installment cycle for investors flagged as 'At-Risk' to boost SIP retention.
2. **Implement Sector Concentration Guardrails:** Cap single-sector portfolio allocations at ≤35% in Large Cap funds to prevent excessive Sector HHI risk spikes.
3. **Embed Risk Appetite Matching Engine:** Integrate the simple recommender engine (recommender.py) into retail mobile apps to guide conservative investors toward Large Cap and Hybrid funds.

16. Project Limitations

- **Historical Time Window:** Daily NAV data spans 4.4 years (2022–2026). Evaluating 10-year macroeconomic market cycles would enhance long-term VaR calibration.
- **Portfolio Holdings Scope:** Detailed sector allocation data (`09\_portfolio\_holdings.csv`) covers 34 equity schemes; underlying bond holdings for debt funds were not included in raw holdings.
- **Power BI Cloud Publishing:** The interactive Power BI dashboard is fully functional in Power BI Desktop mode (`Dashboard.pdf`); online cloud publishing URL depends on organizational tenant permissions.

17. Future Scope & System Enhancements

1. **GARCH Volatility & Monte Carlo VaR:** Upgrade risk modeling from historical percentile methods to dynamic GARCH(1,1) dynamic volatility estimation and Monte Carlo tail simulations.
2. **Live AMFI API Data Pipeline:** Build automated daily web scrapers connecting directly to the AMFI India API for instant daily NAV ingestion.
3. **Machine Learning Churn Prediction:** Train Logistic Regression and XGBoost classifiers on investor transaction gap features to predict SIP cancellation probabilities 60 days in advance.

18. Conclusion

The **Bluestock Mutual Fund Analytics Capstone** delivers a production-ready quantitative analytics and business intelligence infrastructure. By uniting robust data engineering ETL pipelines, relational star schema warehousing, advanced quantitative risk metrics (VaR/CVaR, Rolling Sharpe, HHI), behavioral cohort modeling, and executive Power BI dashboards, this project equips Bluestock Fintech with actionable insights to protect retail capital, reduce SIP churn, and optimize portfolio returns.